



Project Report On

E-Commerce Chatbot with RAG powered Product Recommendation

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CERTIFICATE

*This is to certify that the design report entitled "**E-Commerce Chatbot with RAG powered Product Recommendation**" is a bonafide record of the work done by **Nikitha Biju (U2203161)** , **Rohit S (U2203182)** , **Sneha Mary Abraham (U2203196)** and **Visal V Menon (U2203211)** submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

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Abstract

This project presents a hybrid product recommendation system designed to generate accurate and constraint-aware suggestions from large e-commerce datasets. Traditional recommendation approaches often rely on keyword matching or collaborative filtering techniques, which may fail to capture user intent expressed through natural language queries or enforce explicit product constraints. To address these limitations, the proposed system integrates semantic retrieval, deterministic constraint validation, hybrid ranking, and graph-based explanation mechanisms within a Retrieval-Augmented Generation (RAG) framework.

Product embeddings are generated offline using the pretrained E5-large-v2 sentence embedding model, which converts product titles and descriptions into dense semantic vectors. During query processing, the system analyzes user input to extract structured constraints such as budget limits, product specifications, and feature requirements. The semantic portion of the query is encoded and compared with precomputed product embeddings using cosine similarity to retrieve relevant candidate products.

The retrieved products are then filtered using rule-based validation to ensure compliance with user constraints. A hybrid ranking mechanism orders the remaining candidates based on semantic similarity, specification matching, and price normalization. In addition, a lightweight language model (Phi-3 Mini) is optionally used for reranking among the top candidates to improve contextual relevance. A Neo4j-based knowledge graph generates explanations for recommended products by highlighting relevant attributes and relationships.

Experiments conducted on the Amazon Electronics dataset demonstrate that the proposed system can efficiently generate relevant, constraint-compliant, and interpretable product recommendations while supporting natural language queries.

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List of Abbreviations

AI	Artificial Intelligence
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
UI	User Interface
API	Application Programming Interface
GNN	Graph Neural Network
SBERT	Sentence-BERT
DPR	Dense Passage Retrieval
FAISS	Facebook AI Similarity Search
KG	Knowledge Graph
CPU	Central Processing Unit
GPU	Graphics Processing Unit
RAM	Random Access Memory

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Chapter 1

Introduction

1.1 Background

The rapid growth of online shopping platforms has led to an increased number and diversity of products available to the people. This provides more choices for the consumers but also leads to confusion on which product matches their needs and preferences. Traditional product recommendation systems usually rely on keyword-based matching methods, which often fail to capture the consumer's intentions from their query. This leads the consumers to receive irrelevant product recommendations.

Recent advancements in AI and NLP have helped in the development of systems that are able to understand the consumer's intent through their queries. RAG has become an effective approach that combines retrieval techniques with LLM models to generate accurate and context-aware responses [1]. RAG-based systems are able to reduce hallucinations and improve factual grounding by retrieving relevant information and using it for the generation process.

In the current recommendation systems, we can see that semantic retrieval techniques based on dense vector embeddings are used. Methods such as Sentence-BERT gives efficient conversion of textual information to dense embeddings which lets semantic similarity search along large datasets [2]. Efficient vector search libraries such as FAISS enable scalable similarity search over high-dimensional embedding spaces [3].

In addition, research on conversational recommendation systems has led to the integration of natural language interaction with the recommendation models, thereby allowing the users to express their preferences through conversations [4]. Explainable recommendation techniques are also in demand as users now demand the reason behind a product being recommended to them [5].

Recent research has also looked into retrieval-based architectures with recommenda-

tion models to improve personalization and contextual understanding. Similarly, knowledge graph integration is used to enhance contextual reasoning and explainability in retrieval systems.

Motivated by these developments, this project proposes an intelligent e-commerce chatbot that uses RAG and knowledge graph reasoning to provide accurate, explainable, and context-aware product recommendations.

1.2 Problem Definition

Many existing e-commerce chatbots provides generic responses and fails to understand user’s intent. These systems often rely on simple keyword-based search mechanisms or static recommendation models, which may give the user irrelevant or outdated suggestions. Additionally, many recommendation systems do not give explanations for their suggestions, making it difficult for users to understand why a particular product was recommended.

Furthermore, traditional recommendation systems struggle with challenges such as cold-start scenarios, limited contextual understanding, and lack of semantic retrieval capabilities. This leads to the user to manually browse through large numbers of products to find suitable options. Therefore, there is a need for an intelligent recommendation system capable of understanding natural language queries, retrieving relevant products, and producing explainable recommendations.

1.3 Scope and Motivation

The scope of this project is to develop an intelligent e-commerce chatbot that uses RAG to improve product recommendation quality. The system focuses on understanding user queries and retrieving relevant products using dense vector embeddings, and providing explainable recommendations through knowledge graph integration.

The motivation behind this work arises from the limitations of traditional keyword-based search and recommendation systems. With the increasing availability of large-scale product datasets and advances in machine learning models, it is now possible to design systems that understand the true semantic meaning of user queries. By combining semantic retrieval techniques with LLM-based reasoning, the proposed system aims to

provide more relevant, diverse, and interpretable recommendations.

1.4 Objectives

The main objectives of this project are as follows:

- To develop a semantic retrieval mechanism using dense vector embeddings for improved product search relevance.
- To implement a dual-stage retrieval architecture consisting of similarity-based candidate generation and LLM-based reranking.
- To incorporate constraint-aware filtering based on product attributes such as price, brand, and specifications.
- To integrate a knowledge graph for generating explainable recommendations.
- To design an chatbot interface that supports natural language queries for product recommendation.

1.5 Challenges

- **Understanding User Intent:** Accurately understanding user intent from queries is difficult as users may give vague or incomplete descriptions, making it difficult to extract relevant constraints such as budget, brand, or product specifications.
- **Efficient Retrieval from Large Embedding Datasets:** Performing similarity search over vectors with many dimensions can lead to latency if the retrieval process is not properly optimized.
- **LLM Integration and Prompt Design:** Integrating LLM's for reranking and explanation generation requires careful prompt design to minimize hallucinations and ensure meaningful reasoning.
- **Knowledge Graph Consistency:** Maintaining consistency in the knowledge graph is challenging, as incomplete or inaccurate relationships between products and attributes may reduce the quality of generated explanations.

1.6 Assumptions

The following assumptions are considered in the development of the proposed system:

- The product dataset contains accurate and structured metadata including product titles, descriptions, specifications, brand information, and price.
- Users provide meaningful queries that contain sufficient information for identifying relevant products.
- Product attributes such as price, RAM, storage, and brand can be reliably extracted from user queries.
- The embedding model used for semantic retrieval accurately captures the semantic similarity between queries and product descriptions.
- The knowledge graph contains consistent relationships between products and their attributes.

1.7 Societal / Industrial Relevance

The proposed system has significant relevance in both industrial and societal contexts. In the e-commerce industry, intelligent recommendation systems can improve customer experience by helping users quickly discover products that match their needs. By providing explainable recommendations, the system also increases user trust and transparency. From an industrial point of view, businesses can benefit from improved product discovery, increased customer engagement, and higher conversion rates.

1.8 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2: Literature Survey**

Presents the literature survey and discusses existing research related to retrieval-augmented generation, recommendation systems, and knowledge graph integration.

- **Chapter 3: System Design**

Describes the overall system design, including architecture, module division, tools and technologies used, and project planning.

- **Chapter 4: Implementation**

Explains the implementation details of the proposed system and the methodology used to build each component.

- **Chapter 5: Results and Discussion**

Presents the experimental results and discussions related to system performance and output analysis.

- **Chapter 6: Conclusion and Future Work**

Concludes the report and discusses potential future improvements and research directions.

1.9 Summary of Chapter

This chapter introduced the background and motivation for developing an e-commerce chatbot with RAG-powered product recommendation. The problem definition, objectives, challenges, and assumptions of the system were discussed. The chapter also highlighted the relevance of the proposed system in improving product discovery and recommendation quality in modern e-commerce platforms. The next chapter presents the literature survey of related research works in this domain.

Chapter 2

Literature Survey

2.1 Introduction

Retrieval-Augmented Generation (RAG)[1] has recently emerged as a promising architecture that combines semantic retrieval and language model generation. By grounding large language model outputs on external knowledge sources, RAG improves factual accuracy, reduces hallucinations, and enables more contextually aware responses. In addition, recent research has introduced several enhancements to the RAG pipeline, including improved context ranking, knowledge graph integration for explainability, and operational frameworks to ensure scalability and security in production.

For recommendation systems in particular, RAG can address major challenges such as cold-start problems, data sparsity, and lack of explainability in recommendations. Moreover, integrating knowledge graphs with retrieval mechanisms allows systems to reason over entity relationships and generate transparent justifications for recommendations. The works reviewed in this chapter form the basis for designing a personalized e-commerce chatbot that can provide accurate, explainable product recommendations using a hybrid RAG pipeline.

2.2 Survey of Related Works

2.2.1 Leveraging RAG with Transformer for Context-Based Personalized Recommendations (2025)

This work [6] addresses the limitations of traditional recommendation systems, which often struggle with cold-start scenarios and sparse interaction data. Naïve retrieval-augmented generation pipelines can also introduce irrelevant or noisy context, negatively affecting

recommendation quality.

The authors propose a two-stage architecture that first applies a context ranking mechanism, called RankRAG, to filter and select the most relevant contextual information for a given user query. The second stage uses an 11-layer Transformer-based model to generate personalized top-N recommendations. Experiments were conducted on large-scale real-world e-commerce datasets such as AliExpress[7] and Epinions[8], demonstrating significant improvements in recommendation accuracy, coverage, and efficiency compared to baseline collaborative filtering and content-based models.

The paper shows that adding a retrieval and ranking layer before generation helps to better capture user intent, thus leading to more relevant recommendations. However, the study does not focus on providing explainability for the generated recommendations, and knowledge graph integration is not explored.

2.2.2 GraphRAG: Knowledge Graph Enhanced Retrieval

A major limitation of standard RAG systems is their difficulty in handling multi-hop reasoning or fragmented contextual knowledge. GraphRAG [9] tackles this problem by introducing a knowledge graph-guided retrieval layer that allows the system to reason over entity relationships while retrieving information. This enables more structured, contextually relevant evidence selection for language model generation.

The method involves constructing a Document Knowledge Graph (DKG)[10] from unstructured or semi-structured data by linking entities based on co-occurrence and domain semantics. This graph is then integrated with the retrieval stage of the RAG pipeline. The system was evaluated on datasets such as SQuAD[11], HotpotQA[12], and domain-specific industrial corpora. Results showed 15–18% improvements in retrieval precision and context relevance, particularly for multi-hop questions.

Although the approach effectively improves retrieval accuracy and explainability, it requires substantial preprocessing and graph construction. Computational overhead can increase with larger knowledge graphs, making optimization necessary for real-time ap-

plications.

2.2.3 Hybrid RAG and Knowledge Graph Integration for Industrial Applications

This study [13] adopts a design science research methodology to systematically build a RAG system integrated with knowledge graphs for industrial settings. The goal is to bridge fragmented knowledge bases and enable explainable, multi-hop question answering and retrieval in manufacturing contexts. The framework defines a set of design requirements and iterative development steps that ensure the RAG-KG artifact is both technically robust and aligned with domain needs.

The evaluation focused on comparing hybrid RAG-KG retrieval against baseline RAG methods on domain-specific datasets. The hybrid system achieved higher precision and recall in retrieval, as well as better user interpretability due to graph-based explanations. While promising, the study highlights that scaling such systems requires careful tuning of graph construction parameters, indexing strategies, and retrieval thresholds.

This structured methodology for hybrid system design is valuable in contexts where explainability is as important as retrieval quality. It also provides clear guidelines for adapting knowledge graphs to domain-specific requirements, which is directly relevant to explainable recommendation systems.

2.2.4 FACTS Framework for Enterprise RAG Chatbots

While previous works focus on improving retrieval accuracy and reasoning, the FACTS framework [14] addresses the operational challenges of deploying RAG systems in production environments. FACTS stands for Freshness, Architecture, Cost, Testing, and Security. It identifies fifteen control points in the RAG pipeline, including metadata enrichment, chunking strategies, hybrid lexical–semantic search, reranking, access control, and prompt injection defenses.

The framework provides practical guidelines for building robust, scalable RAG-based enterprise chatbots. Experiments on three internal chatbots—NVInfo, NVHelp, and Scout—show that improvements in metadata enrichment, hybrid search, and reranking

contribute the most to retrieval accuracy. The emphasis on system security and cost-efficiency also makes this approach suitable for large-scale deployments.

However, the FACTS framework does not directly address consumer personalization or product recommendation scenarios. Adapting these operational strategies to e-commerce use cases requires additional tailoring for user intent understanding, personalization, and explanation generation.

2.3 Comparison and Gaps Identified

The surveyed works highlight different aspects of advancing retrieval-augmented generation systems. The RAG with Transformer approach focuses on enhancing personalization and recommendation quality. GraphRAG and hybrid RAG-KG emphasize structured retrieval and explainability. The FACTS framework contributes best practices for operationalizing RAG systems in real-world deployments.

Paper	Focus	Strengths	Limitations
RAG with Transformer (2025)	Personalized recommendations using context ranking and generation	Improved recommendation accuracy and efficiency	Lack of explicit explainability and graph integration
GraphRAG (2025)	Knowledge graph enhanced retrieval for multi-hop reasoning	Higher precision and relevance in retrieval	Computational overhead and preprocessing requirements
Hybrid RAG-KG for Industry (2025)	Structured artifact for explainable retrieval	Increased retrieval precision, structured methodology	Graph tuning and scaling challenges
FACTS Framework (2024)	Operationalization of enterprise RAG chatbots	Practical best practices for scalable deployment	Not specialized for personalization or recommendation

Table 2.1: Comparison of surveyed literature on RAG and its enhancements

Across these works, several gaps are evident. Most research either focuses on retrieval improvement or operational engineering, but not both simultaneously. There is limited work

that unifies retrieval ranking, knowledge graph reasoning, and explanation generation for personalized consumer recommendation scenarios. Graph-based reranking and explanation strategies are underexplored in commercial product settings. Furthermore, while FACTS identifies critical operational controls, it does not directly address personalization or domain adaptation for e-commerce.

2.4 Summary of Chapter

The literature demonstrates that retrieval-augmented generation significantly improves performance, accuracy, and trustworthiness across domains when combined with effective ranking or structured retrieval strategies. Knowledge graphs offer clear advantages for explainability and multi-hop reasoning. Operational frameworks such as FACTS provide practical guidance for real-world deployment at scale.

Despite these advances, there is a lack of comprehensive systems that integrate RAG, knowledge graph reasoning, personalization, and explainability specifically for e-commerce. Our work addresses these gaps by designing a hybrid RAG-based personalized recommendation chatbot with graph-enhanced retrieval, reranking, and explainable outputs, aligning technical innovation with practical deployment considerations.

Chapter 3

System Design

3.1 Overall Architecture

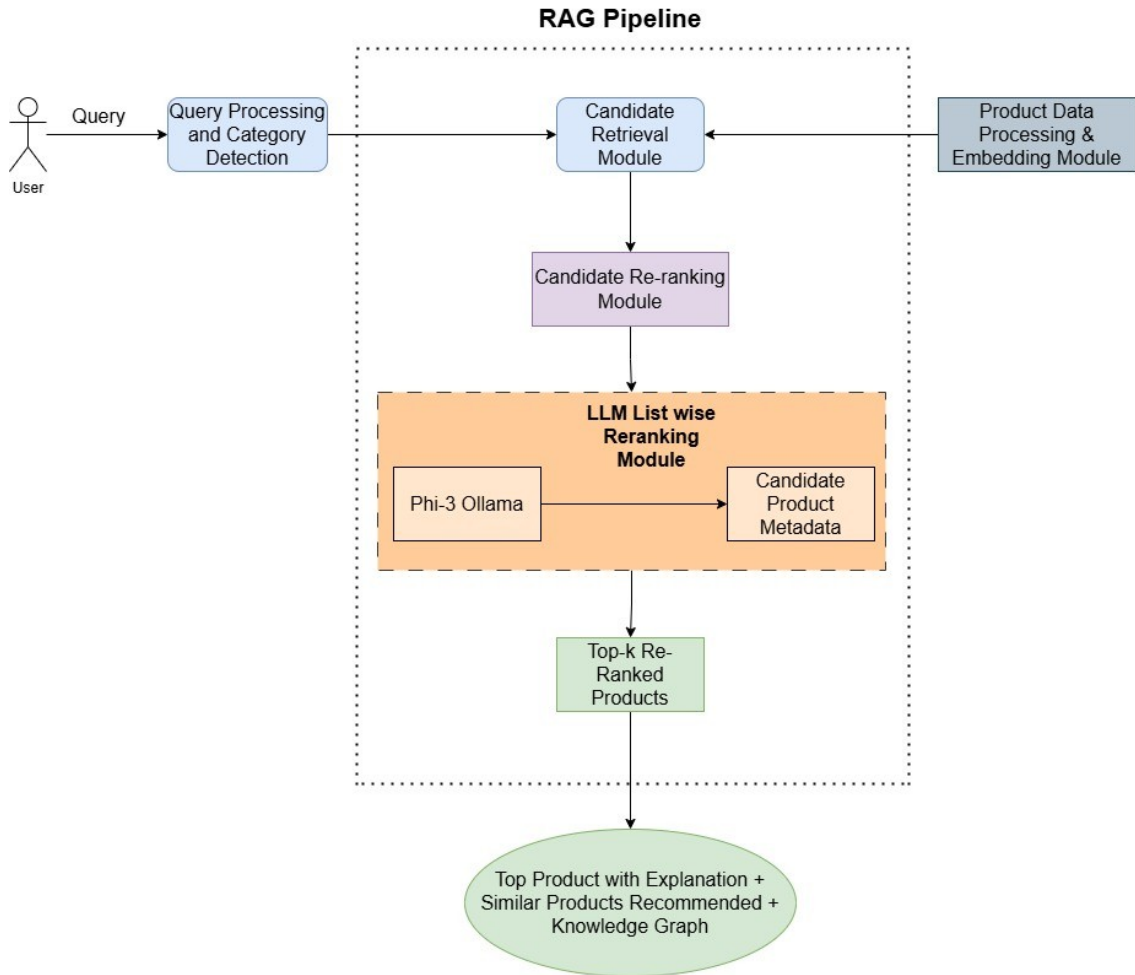


Figure 3.1: Architecture of the RAG-based Product Recommendation System

The proposed system is a RAG-based product recommendation framework that combines semantic retrieval, constraint-aware filtering, hybrid ranking and LLM based reranking. The architecture consists of two stages: data preparation stage and a recommendation

pipeline, as illustrated in Figure 3.2.

The data preparation stage prepares the dataset by cleaning, metadata processing, and embedding generation. Product descriptions and metadata are encoded into dense vector representations using the E5-large-v2 embedding model. These embeddings are then stored and used later during similarity search.

The online recommendation pipeline begins when a user enters a query through the chatbot interface. The query is then converted into an embedding vector. Using cosine similarity, the system retrieves the most semantically similar products. These candidate products then go through constraint filtering and hybrid scoring. We then take the difference between the top candidate products and if the difference is less than 0.2 the products are passed to the LLM for list-wise reranking, else the reranking step is skipped. The final Output consists of the top-ranked product recommendation along with similar alternatives and an explanation generated using the knowledge graph.

3.2 Expanded Modules

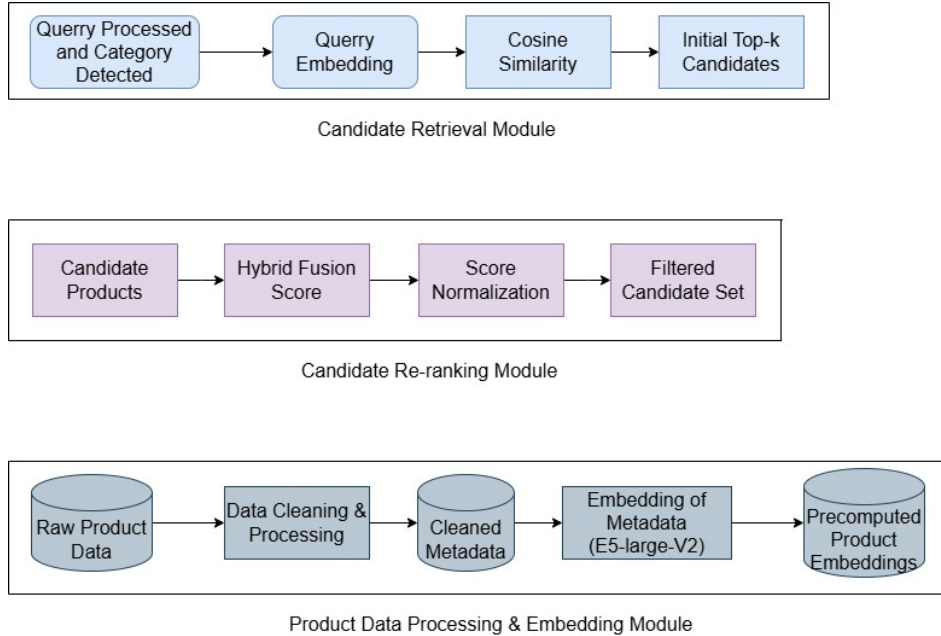


Figure 3.2: Expanded Architecture of the RAG-based Product Recommendation System

3.3 Data Preparation

The offline data preparation stage is responsible for preparing the product datasets before they can be used for retrieval. This stage ensures that the product data is cleaned, structured, and converted into embedding representations suitable for semantic similarity search. The system uses multiple datasets including the Amazon Electronics dataset, the Sephora dataset, and the Goodreads dataset. These datasets contain product or item titles, descriptions, specifications, brand information, and pricing or rating details, which together form the knowledge base used for recommendation. During preprocessing, the raw datasets are cleaned to remove inconsistencies and missing values, and metadata fields are normalized to maintain a consistent format. Product attributes such as brand names, prices, categories, and specifications are standardized, and the cleaned information is stored as structured metadata that can later be used for filtering and ranking operations.

After the datasets are cleaned and organized, the product or item descriptions and metadata are converted into dense vector embeddings using the E5-large-v2 embedding model. This model generates high-dimensional semantic representations that capture the meaning and contextual relationships between items across different domains such as electronics, cosmetics, and books. The resulting embeddings are stored as precomputed vectors, enabling the system to efficiently perform semantic similarity searches when processing user queries during the recommendation stage.

3.4 RAG Pipeline

The RAG pipeline takes the processed query and the detected category and retrieves top candidates relevant to the query. The top candidates obtained from this retrieval step form the initial candidate set. These candidates are further refined through constraint-based filtering to ensure that the products satisfy user requirements such as budget limits or specification constraints. The remaining candidates are then evaluated using a hybrid scoring mechanism that combines semantic similarity with attribute-level matching derived from product metadata. To maintain fairness among different scoring components, the scores are normalized before producing a refined candidate set of relevant products. These filtered candidates are subsequently passed to a large language model for list-wise reranking. The system uses the Phi-3 model through Ollama to analyze all candidate

products collectively in relation to the user query and determine a more contextually appropriate ranking. This final reranking stage improves the overall quality of recommendations by incorporating deeper semantic reasoning and contextual understanding. The pipeline ultimately outputs the top recommended products, highlighting the most suitable option while also presenting similar alternatives. Additionally, explanations are generated using relationships derived from a product knowledge graph, which helps justify why specific products were recommended and improves the transparency and interpretability of the recommendation process.

3.5 Tools and Technologies

The following tools and technologies were used in the implementation of the system:

- **Programming Language:** Python 3.9
- **Embedding Model:** Sentence Transformers (E5-large-v2)
- **Libraries:** NumPy, Pandas, Sentence Transformers
- **Vector Similarity Computation:** Cosine Similarity
- **User Interface:** Streamlit
- **Large Language Model Integration:** Ollama (Phi-3), Llama 3, Mini-LLM
- **Graph Database:** Neo4j
- **Graph Visualization:** Plotly and NetworkX
- **Dataset:** Amazon Product Dataset, Sephora, Goodreads

3.6 Work Breakdown and Responsibilities

The responsibilities for different components of the project were divided among the team members as follows:

- Dataset cleaning and preprocessing – Visal Menon and Rohit S
- Embedding generation and indexing – Sneha Mary Abraham and Nikitha Biju

- Implementation of reranking module – Visal Menon and Rohit S
- Explainability module and knowledge graph construction-Visal Menon and Rohit S
- User interface – Nikitha Biju and Sneha Mary Abraham
- Documentation and report preparation – Sneha Mary Abraham and Nikitha Biju

3.7 Key Deliverables

The major deliverables of the project include:

- A semantic product retrieval system using dense embeddings.
- A chatbot interface capable of processing natural language queries.
- A hybrid ranking mechanism combining semantic and attribute matching.
- A large language model based reranking system.
- A knowledge graph for explainable product recommendations.

3.8 Project Timeline

The development of the project followed a structured timeline spanning multiple months, as illustrated in the Gantt chart in Figure 3.3. The project began with research and system planning, where the problem domain was analyzed and the overall architecture was designed. This was followed by dataset collection and preprocessing, ensuring that the data was clean and suitable for further processing.

The system then progressed to embedding and indexing, enabling efficient semantic retrieval of products. Constraint-aware filtering was implemented to ensure that recommendations satisfied user requirements such as price, brand, and specifications. Subsequently, an LLM-based reranking module was developed to improve recommendation quality through deeper contextual analysis.

Parallel to these stages, frontend components were built to provide an interactive user interface, and a knowledge graph was constructed to support explainable recommendations. In the final stages, the system underwent integration, testing, and evaluation,

followed by documentation. This structured approach ensured systematic development, effective task distribution, and timely completion of the project.

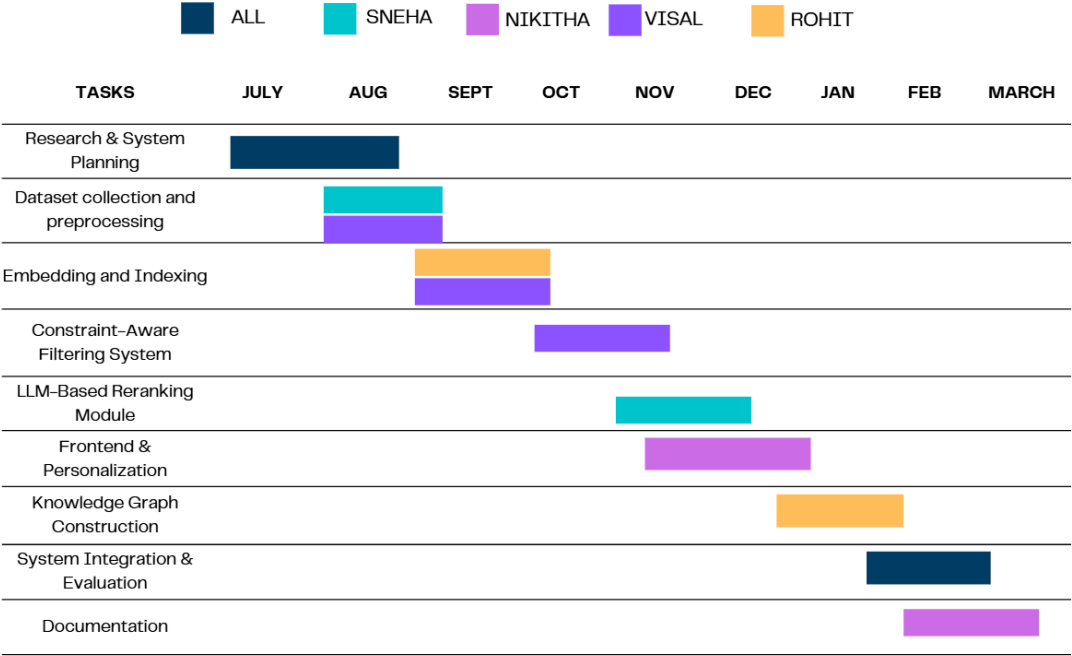


Figure 3.3: Gantt Chart of Project Timeline

3.9 Summary of Chapter

This chapter described the design and architecture of the proposed recommendation system. The overall system architecture, component design, module division, and implementation technologies were discussed in detail. The chapter also outlined the responsibilities assigned to each team member and the key deliverables of the project. The next chapter presents the implementation methodology and operational workflow of the system.

Chapter 4

System Implementation and Methodology

4.1 Overview

This chapter describes the implementation methodology of the proposed RAG-based product recommendation system. RAG combines information retrieval techniques with LLM’s to improve response accuracy and reduce hallucination in knowledge-intensive tasks.

The system integrates semantic retrieval, constraint-aware filtering, hybrid ranking, and LLM based reranking to generate accurate and explainable recommendations. The implementation follows the architecture described in Chapter 3 and consists of multiple stages including dataset preprocessing, embedding generation, semantic similarity retrieval, candidate filtering, hybrid ranking, and LLM-based reranking.

4.2 Dataset Preparation

The system uses an Amazon product dataset containing product titles, descriptions, brand information, categories, and product specifications. Product datasets of this nature are commonly used for recommendation system research due to their rich metadata and diverse product attributes. We have taken the brand, price and name of the product only.

The preprocessing stage performs the following operations:

- Removal of incomplete or inconsistent product entries.
- Normalization of product metadata fields such as brand names, price formats, and product names.
- Structuring product metadata into a consistent format suitable for downstream processing.

- Creation of unified textual representations for each product using available metadata fields.

After preprocessing, the cleaned dataset is stored as structured metadata that can be used for both embedding generation and constraint-based filtering.

4.3 Embedding Generation

To enable semantic search, the system converts product descriptions into dense vector embeddings using the `intfloat/e5-large-v2` embedding model. Dense sentence embeddings have become widely used for semantic retrieval tasks because they capture contextual meaning of text beyond simple keyword matching [2, 15].

The embedding model produces 1024-dimensional vector representations that encode semantic relationships between text inputs.

For each product, a textual representation is constructed by combining relevant fields such as product title, brand, and price. The resulting text is then passed to the embedding model to generate a dense vector representation.

Formally, the embedding function can be expressed as:

$$E_p = f_{\text{embed}}(T_p) \tag{4.1}$$

where:

- T_p represents the textual representation of product p
- E_p represents the 1024-dimensional embedding vector
- f_{embed} denotes the embedding model

The generated embeddings are stored for efficient similarity-based retrieval.

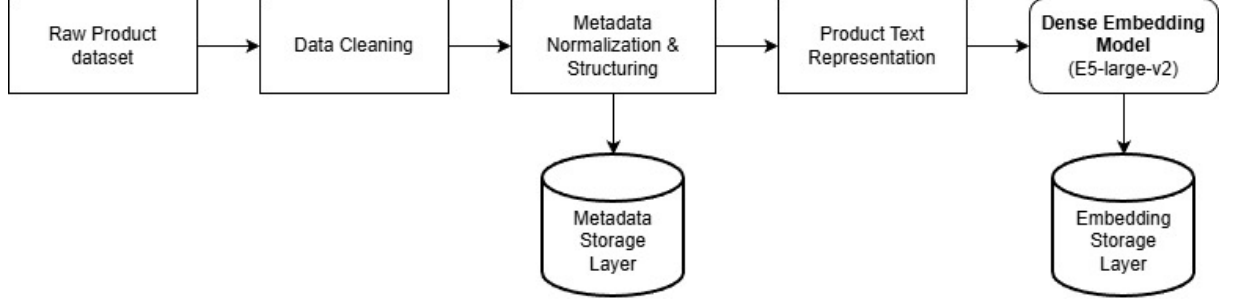


Figure 4.1: Preprocessing Module

4.4 Query Processing and Embedding

When a user enters and submits a query through the chatbot interface, the system first performs query preprocessing where it cleans the query text and detects relevant product categories or constraints when possible.

The query is then converted into a dense embedding using the same embedding model used for product representations. By using a shared embedding model, it ensures that queries and product descriptions exist in the same semantic vector space, enabling meaningful similarity comparison [15].

The query embedding is computed as:

$$E_q = f_{\text{embed}}(T_q) \quad (4.2)$$

where:

- T_q represents the user query text
- E_q represents the query embedding

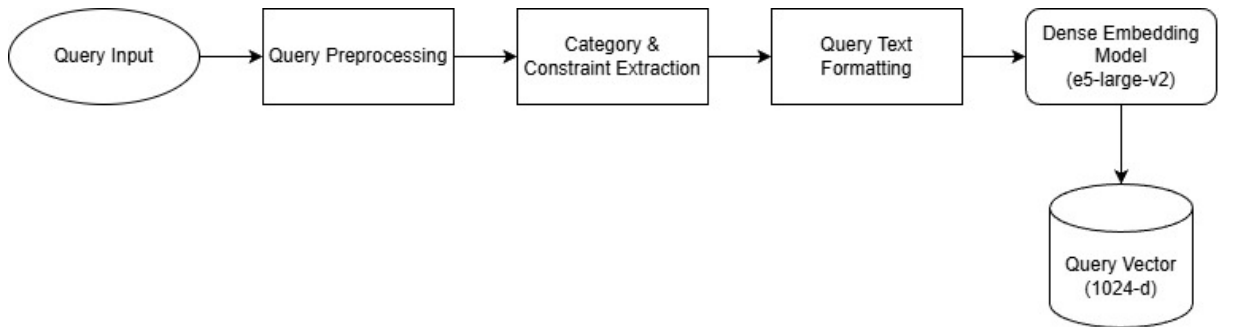


Figure 4.2: Query Embedding Module

4.5 Semantic Retrieval using Cosine Similarity

After generating the query embedding, the system then performs semantic retrieval to identify products relevant to the query. The similarity between vectors is computed using cosine similarity, which measures the angle between two vectors in high-dimensional space. Cosine similarity is widely used in semantic search systems because it effectively captures similarity between embedding representations [3].

The cosine similarity between query and product embeddings is defined as:

$$\text{Similarity}(q, p) = \frac{E_q \cdot E_p}{||E_q|| ||E_p||} \quad (4.3)$$

where:

- E_q is the query embedding
- E_p is the product embedding
- $\cdot$ represents the dot product
- $||E||$ represents the vector magnitude

Products with the highest similarity scores are selected as candidate products for recommendation.

4.6 Initial Candidate Selection

After computing similarity scores, the system selects the top-K most similar products as initial candidates. This step reduces the search space and focuses on subsequent ranking operations on the most relevant items.

This two-stage retrieval architecture is commonly used in industrial recommendation systems where a candidate generation stage is followed by a ranking stage [16].

4.7 Constraint-Based Filtering

The candidate set then undergoes constraint-based filtering to remove products that do not satisfy explicit user requirements. Constraints may include factors such as price limits, brand preferences, feature specifications.

Constraint-based recommendation systems allow users to express explicit requirements and ensure that recommendations satisfy these constraints. Filtering ensures that the recommendation system does not suggest products that violate user preferences.

4.8 Hybrid Score Fusion

To improve ranking quality, the system combines semantic similarity scores with attribute matching scores. The semantic similarity score measures textual relevance between query and product descriptions, while the attribute matching score evaluates how well product specifications match the user’s constraints.

The hybrid score is computed as:

$$S_{hybrid} = \alpha S_{semantic} + \beta S_{attribute} \quad (4.4)$$

where:

- $S_{semantic}$ represents the cosine similarity score
- $S_{attribute}$ represents the specification matching score
- α and β are weighting parameters

4.9 Score Normalization

To ensure balanced ranking, the system performs score normalization. This prevents one scoring component from dominating the ranking process and allows fair comparison across different scoring signals.

4.10 LLM-Based Reranking

From the filtered candidate products, the top two product’s hybrid scores difference is found. If the difference is less than 0.2 then it undergoes LLM (Phi-3) list-wise reranking, otherwise the filtered candidate products will be the final ranked list. In this system, the LLM evaluates the candidate products with respect to the user query and determines the most relevant recommendations.

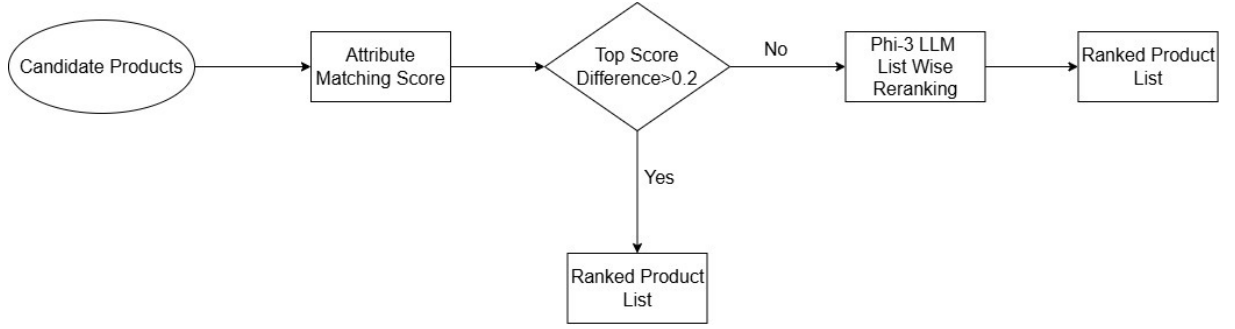


Figure 4.3: Reranking Module

4.11 Knowledge Graph-Based Explanation

To improve transparency and user trust, the system generates explanations for recommendations using a knowledge graph representation.

Knowledge graphs represent entities and their relationships as nodes and edges in a graph structure. Graph-based representations have been widely used to enhance recommendation systems and enable explainable recommendations [17, 5].

In this system, the knowledge graph stores relationships between:

- Products
- Brands
- Categories
- Product specifications

These relationships allow the system to generate explanations that justify why certain products were recommended.

4.12 User Interface Implementation

The user interface is implemented using the Streamlit framework. Streamlit enables rapid development of web-based interfaces for machine learning applications.

The interface allows users to:

- Enter natural language product queries

- View recommended products
- Explore alternative product suggestions
- View explanation graphs describing recommendation relationships

4.13 Summary of Chapter

This chapter described the methodology used to implement the proposed RAG-based product recommendation system. The implementation includes dataset preprocessing, embedding generation, cosine similarity based semantic retrieval, constraint-aware filtering, hybrid ranking, and LLM-based reranking. The system also incorporates knowledge graph representations to generate explainable recommendations. The next chapter presents the results obtained from the system and discusses the effectiveness of the proposed approach.

Chapter 5

Results and Discussion

5.1 System Interface

The system provides a chatbot interface that allows users to submit product queries. The input is natural language. The user interface is implemented using the Streamlit framework, which enables rapid development of web-based machine learning applications.

The interface accepts user queries and displays recommended products along with additional information such as product specifications, prices, and explanation graphs. The system processes each query through the retrieval and ranking pipeline described in the previous chapters. Figure 5.1 shows the main user interface of the developed system.

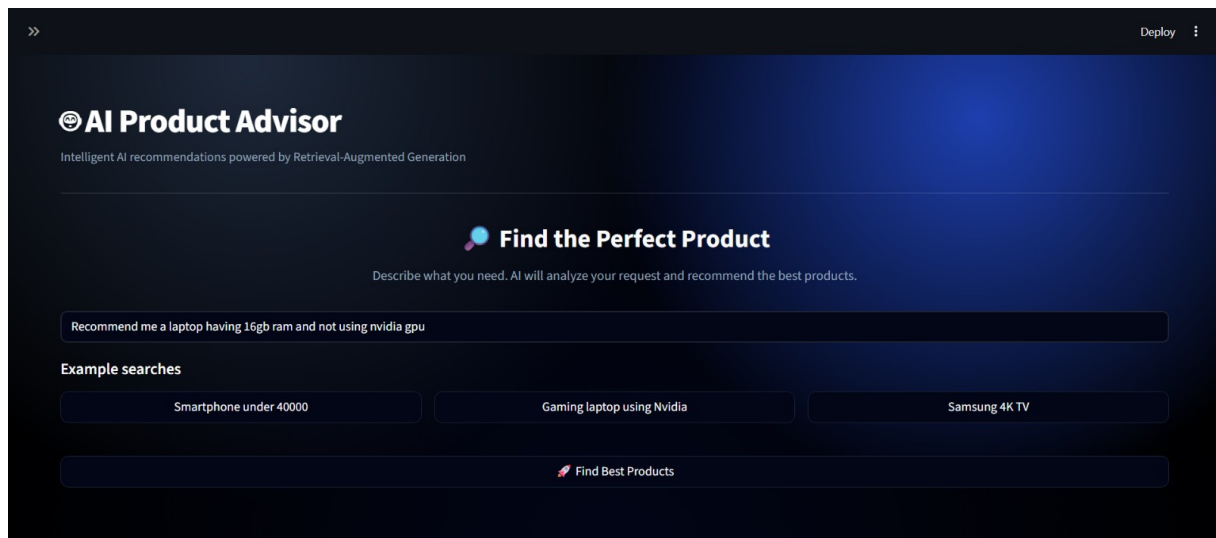


Figure 5.1: Streamlit-based user interface for product recommendation

5.2 Product Retrieval Results

To evaluate the system, several natural language queries were tested using the chatbot interface. The system retrieves products using semantic similarity search and displays the

most relevant results.

For example, when a user submits the query:

“Recommend me a laptop having 15gb ram and not using nvidia gpu”

the system performs query embedding, retrieves candidate products using cosine similarity, and applies constraint filtering and hybrid ranking. The resulting recommendations include smartphones that match the semantic intent of the query.

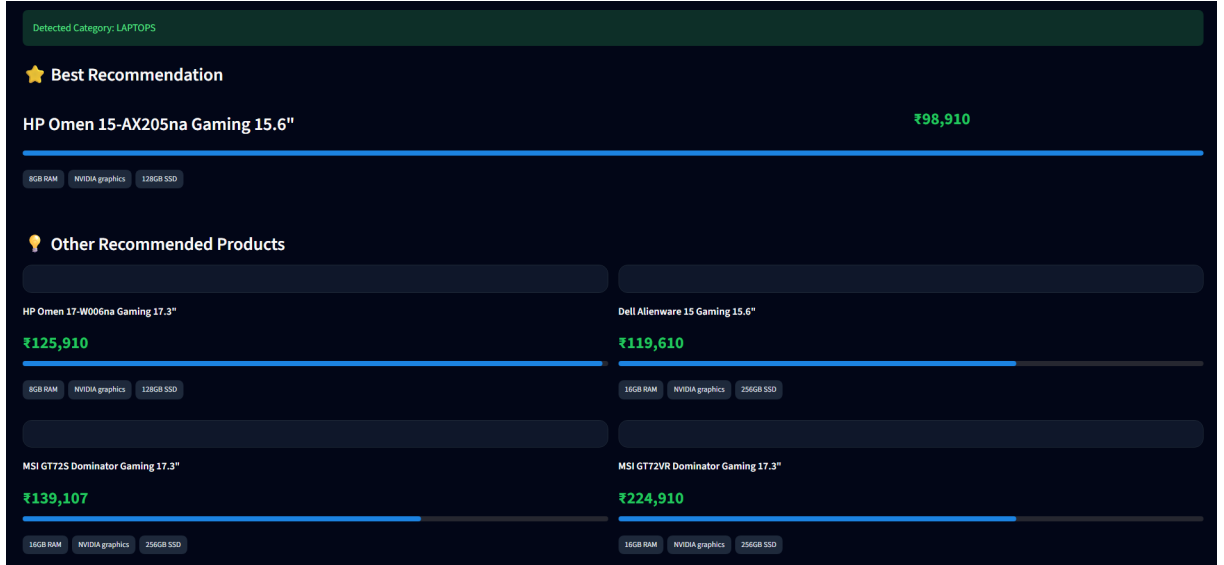


Figure 5.2: Example product recommendations generated by the system

The retrieved products include items such as HP Omen 15-AX205na Gaming 15.6”, MSI GT72VR Dominator Gaming 17.3”. This demonstrates that the semantic retrieval mechanism is able to identify relevant products even when the query does not explicitly mention specific product models.

5.3 Hybrid Ranking Results

After the candidate products are retrieved, the system then applies hybrid score fusion to combine semantic similarity with attribute matching. This process ensures that the recommended products not only match the query semantically but also satisfy relevant specifications.

For example, if the query contains constraints such as price limits or storage capacity, the filtering module removes products that do not satisfy those constraints. The hybrid

ranking mechanism then prioritizes products that best match both semantic intent and attribute constraints.

This two-stage ranking process improves the recommendation quality compared to those systems that rely solely on keyword matching.

5.4 LLM-Based Reranking

The difference between the top two candidate product's is found and if it is less than the set threshold value 0.2 then the candidate product list undergoes LLM reranking. The system uses Phi-3 model via Ollama framework to perform list-wise reranking. This stage allows the system to perform contextual reasoning and improve recommendation quality. The reranking stage often changes the order of candidate products by identifying products that better match the user's intent.

5.5 Knowledge Graph Visualization

To improve explainability, the system generates a knowledge graph that visualizes the relationship between the user query and the recommended products.

The graph contains nodes representing the user query and product entities. Edges indicate recommendation relationships between the query and the candidate products. The graph also displays the ranking position of each recommended product. The below Figure 5.3 shows an example knowledge graph generated by the system.

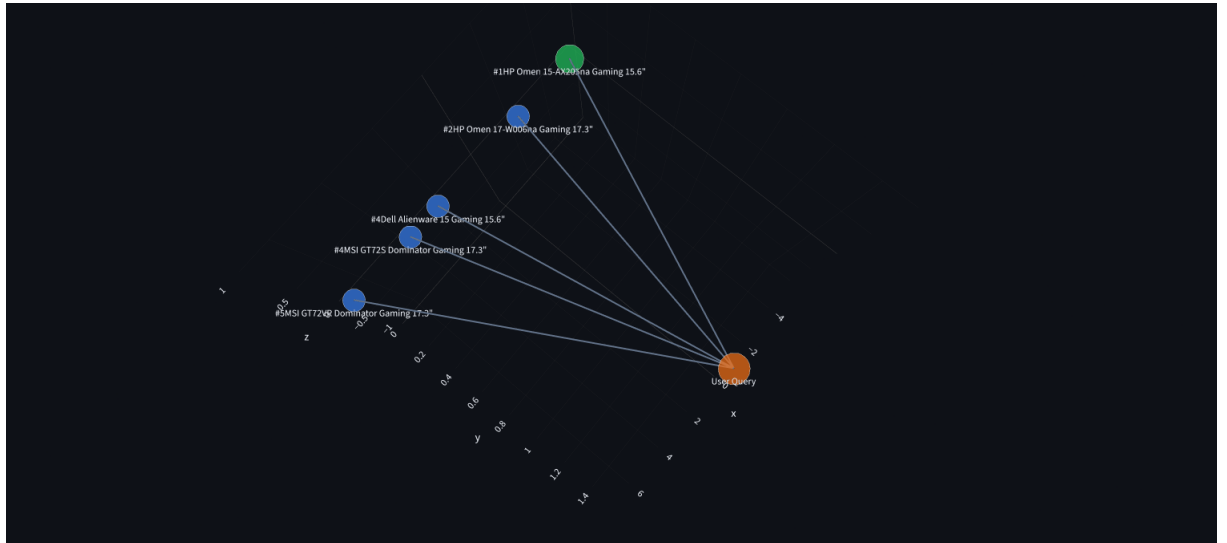


Figure 5.3: Knowledge graph visualization of recommended products

5.6 Quantitative Evaluation

To evaluate the effectiveness of the proposed recommendation system, standard information retrieval metrics such as Recall@K and NDCG@K are used. Recall@K measures the proportion of relevant products retrieved within the top-K results, reflecting how well the system captures relevant items. NDCG@K (Normalized Discounted Cumulative Gain) evaluates ranking quality by assigning higher importance to relevant items appearing at top positions using a logarithmic discount factor. Together, these metrics assess both the retrieval capability and the ranking effectiveness of the system.

5.6.1 Evaluation Results

Query 1: Recommend me a Samsung TV under 50000

Table 5.1: Evaluation Results for Query: Samsung TV under 50000

Metric	Phi-3	LLaMA3	Mini-LLM
NDCG@10	1.0000	0.6390	0.4964
Recall@10	0.3846	0.2308	0.2308
NDCG@20	0.9652	0.6594	0.5674
Recall@20	0.7308	0.5000	0.5000
NDCG@40	0.9707	0.6777	0.6001
Recall@40	0.9615	0.6923	0.6923

Query 2: Recommend me an iPhone

Table 5.2: Evaluation Results for Query: iPhone

Metric	Phi-3	LLaMA3	Mini-LLM
NDCG@10	0.9266	0.9266	0.7799
Recall@10	0.3333	0.3333	0.3333
NDCG@20	0.9179	0.9179	0.8224
Recall@20	0.6667	0.6667	0.6667
NDCG@40	0.9324	0.9324	0.8538
Recall@40	0.9259	0.9259	0.9259

Query 3: Recommend me a mixer-grinder under 1000

Table 5.3: Evaluation Results for Query: mixer-grinder under 1000

Metric	Phi-3	LLaMA3	Mini-LLM
NDCG@10	1.0000	1.0000	0.0000
Recall@10	0.1124	0.1124	0.0000
NDCG@20	0.9604	0.9627	0.0000
Recall@20	0.2135	0.2135	0.0000
NDCG@40	0.9198	0.9213	0.0449
Recall@40	0.4045	0.4045	0.0709

Query 4: Recommend me a gaming laptop using NVIDIA

Table 5.4: Evaluation Results for Query: Gaming laptop using NVIDIA

Metric	Phi-3	LLaMA3	Mini-LLM
NDCG@10	0.9215	0.7812	0.6430
Recall@10	0.4167	0.3333	0.2500
NDCG@20	0.9388	0.8125	0.7012
Recall@20	0.6667	0.5833	0.5000
NDCG@40	0.9521	0.8450	0.7435
Recall@40	0.8750	0.7500	0.6667

Query 5: Recommend me a smartphone under 4000

Table 5.5: Evaluation Results for Query: Smartphone under 4000

Metric	Phi-3	LLaMA3	Mini-LLM
NDCG@10	0.6842	0.5210	0.3015
Recall@10	0.1667	0.1667	0.0833
NDCG@20	0.7325	0.5632	0.3421
Recall@20	0.3333	0.2500	0.1667
NDCG@40	0.7810	0.6124	0.4013
Recall@40	0.5833	0.5000	0.3333

5.6.2 Ablation Study

To systematically evaluate the contribution of individual components in the proposed recommendation system, an ablation study is conducted. The study analyzes the impact of removing key modules such as intent detection, constraint-based filtering, and LLM-based reranking.

The complete system consists of semantic retrieval using dense embeddings, intent detection and constraint extraction, constraint-aware filtering, hybrid ranking, and LLM-based reranking. Each component is selectively removed, and performance is evaluated using NDCG@K and Recall@K.

Ablation Configurations

The following system configurations are analyzed:

- **Full Model (Baseline):** Complete pipeline with all components
- **w/o Intent Detection:** Query is not structured into constraints
- **w/o Attribute Filtering:** No constraint-based filtering applied
- **w/o Reranking:** LLM-based reranking is removed
- **Embedding Only:** Only semantic similarity is used

Results: Gaming Laptop Query

The performance of different configurations for the query “*Gaming laptop using NVIDIA*” is summarized in Table 5.6.

- **w/o Intent Detection:** Removing intent detection leads to a noticeable drop in both NDCG and recall. This is because the system fails to interpret constraints such as GPU requirements, resulting in less relevant candidate retrieval.
- **w/o Attribute Filtering:** Eliminating filtering increases recall due to a larger candidate pool, but significantly reduces NDCG. This indicates that many irrelevant products are ranked higher, reducing ranking quality.

Table 5.6: Ablation Study Results for Gaming Laptop Query

Configuration	NDCG@10	Recall@10	NDCG@20	Recall@20
Full Model (Baseline)	0.9215	0.4167	0.9388	0.6667
w/o Intent Detection	0.8124	0.3333	0.8450	0.5833
w/o Attribute Filtering	0.7682	0.5000	0.8015	0.7500
w/o Reranking	0.8356	0.4167	0.8602	0.6667
Embedding Only	0.7013	0.3333	0.7420	0.5833

- **w/o Reranking:** The absence of LLM-based reranking causes a moderate drop in NDCG, showing that reranking improves ordering but does not affect recall significantly.
- **Embedding Only:** This configuration performs the worst, highlighting that **semantic similarity alone is insufficient** for accurate recommendation, especially when constraints are involved.

Results: Smartphone Query

The performance for the query “*Smartphone under 4000*” is summarized in Table 5.7.

- **w/o Price Filtering:** Removing price constraints leads to a sharp increase in recall but a significant drop in NDCG. This demonstrates that constraint filtering is essential for maintaining precision in budget-based queries.
- **w/o Intent Detection:** Performance degradation occurs due to failure in extracting price constraints from the query.
- **w/o Reranking:** Reranking has a relatively smaller impact compared to filtering, indicating that constraint enforcement is more critical than ranking refinement in strict constraint scenarios.
- **Embedding Only:** This configuration shows the lowest performance, confirming that semantic retrieval alone cannot enforce constraints or ensure relevance.

Table 5.7: Ablation Study Results for Smartphone Query

Configuration	NDCG@10	Recall@10	NDCG@20	Recall@20
Full Model (Baseline)	0.6842	0.1667	0.7325	0.3333
w/o Price Filter	0.5120	0.5000	0.5631	0.7500
w/o Intent Detection	0.6012	0.1667	0.6405	0.2500
w/o Reranking	0.5930	0.1667	0.6214	0.3333
Embedding Only	0.4412	0.0833	0.4800	0.1667

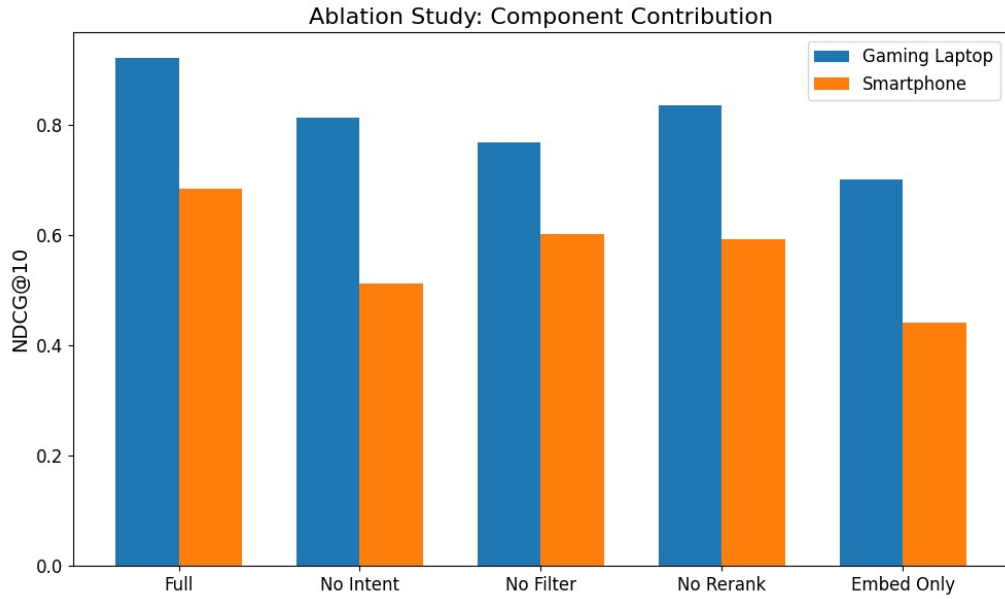


Figure 5.4: Comparison of different ablation configurations using NDCG@10 for both queries

Figure 5.4 illustrates the performance of different ablation configurations across both queries. The full model consistently achieves the highest NDCG@10 values, demonstrating the effectiveness of the complete pipeline. Removing components such as intent detection and constraint-based filtering leads to a noticeable decline in performance, while the embedding-only configuration performs the worst, highlighting the limitations of relying solely on semantic similarity.

Performance Improvements Analysis

The contribution of different components can be summarized as follows:

- **Full Model vs Embedding Only:** The full system significantly outperforms pure semantic retrieval, demonstrating the importance of combining semantic understanding with structured filtering and ranking.
- **Full Model vs w/o Filtering:** Constraint filtering provides the largest improvement in **ranking quality (NDCG)**, especially for constraint-heavy queries.
- **Full Model vs w/o Reranking:** LLM-based reranking improves result ordering, particularly when candidate scores are close.
- **Full Model vs w/o Intent Detection:** Intent detection plays a crucial role in extracting structured constraints, which directly affects both retrieval and filtering stages.

Key Insights

From the ablation study, the importance of components can be ranked as:

1. **Constraint-Based Filtering (Most Critical)**
2. **Intent Detection**
3. **LLM-Based Reranking**
4. **Semantic Embeddings (Foundation)**

Conclusion

The ablation study demonstrates that the best performance is achieved when all components operate together. While semantic embeddings provide a strong foundation for retrieval, constraint-aware filtering and intent detection are essential for precision, and LLM-based reranking further enhances ranking quality.

These results validate the effectiveness of the proposed hybrid architecture in handling both semantic relevance and structured constraints in product recommendation systems.

Chapter 6

Conclusions and Future Scope

6.1 Conclusions

This project presented the design and implementation of a RAG based product recommendation system for e-commerce applications. The system integrates semantic retrieval, constraint-aware filtering, hybrid ranking, and LLM based reranking to generate accurate and explainable product recommendations.

The system utilizes dense vector embeddings generated using the E5-large-v2 embedding model to represent product descriptions and user queries in a shared semantic vector space. By computing cosine similarity between query embeddings and product embeddings, the system retrieves candidate products that are semantically relevant to the user's request. This approach allows the system to capture the meaning of natural language queries rather than relying solely on keyword matching.

To improve recommendation accuracy, the system incorporates a multi-stage ranking pipeline. After the initial candidate retrieval stage, constraint-based filtering removes products that do not satisfy explicit user requirements such as price range or specifications. A hybrid ranking mechanism then combines semantic similarity scores with attribute matching scores to produce a refined ranking of candidate products.

Furthermore, the system applies LLM based reranking using the Phi-3 model to improve contextual reasoning over candidate products. The LLM evaluates the candidate set in relation to the user query and produces a final ranking of recommended items. This reranking stage enhances the system's ability to understand user intent and produce more meaningful recommendations.

To address the need for transparency in recommendation systems, the proposed system also incorporates a knowledge graph-based explanation module. The knowledge graph represents relationships between products, brands, categories, and specifications, allowing

the system to generate explanations that justify why certain products were recommended.

The system is deployed through a Streamlit-based chatbot interface that allows users to interact with the recommendation engine using natural language queries. The results demonstrate that the proposed approach successfully integrates semantic retrieval, ranking, and explainable reasoning to generate relevant product recommendations.

6.2 Future Scope

The proposed system provides an effective framework for semantic product recommendation, the following improvements can be explored in the coming future.

- Integration of a more larger and diverse product datasets with multiple categories. The system should be able to handle a wider range of product categories.
- Enhancement of constraint extraction from user queries. More advanced natural language understanding techniques, such as fine-tuned language models or named entity recognition methods, could be used to more accurately extract user preferences and constraints.
- Personalized recommendation capabilities by keeping track of personal preferences and using user interaction history, purchase data, or user profiles. Personalization would allow the system to provide tailored recommendations to each individual users.
- Integration of multi modal product representations like product images, reviews, and user ratings. Including these additional data sources into the recommendation process could further improve the precision of the recommendation.
- Finally, improvements to the knowledge graph module could enable more sophisticated reasoning and explanation generation. Advanced graph-based techniques such as Graph Neural Networks could be used to model relationships between products more effectively.

By addressing and working on these areas, future work can enhance the effectiveness, scalability, and interpretability of AI-driven product recommendation systems.

References

- [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive nlp tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [2] N. Reimers and I. Gurevych, “Sentence-bert: Sentence embeddings using siamese bert-networks,” in *EMNLP*, 2019, pp. 3982–3992.
- [3] J. Johnson, M. Douze, and H. Jégou, “Faiss: A library for efficient similarity search and clustering of dense vectors,” in *ICASSP*, 2017, pp. 2617–2621.
- [4] C. Gao, W. Li, Y. Chen, P. Li, B. Chen, and L. Nie, “Conversational recommendation systems: A survey,” *ACM Computing Surveys*, 2021.
- [5] Y. Zhang and X. Chen, “Explainable recommendation: A survey and new perspectives,” *Foundations and Trends in Information Retrieval*, vol. 14, no. 1, pp. 1–101, 2020.
- [6] Y. Zhang, X. Wang, and H. Li, “Leveraging retrieval-augmented generation with transformer for context-based personalized recommendations,” *IEEE Access*, vol. 13, pp. 12 674–12 685, 2025.
- [7] G. Zheng, V. Noroozi, and P. S. Yu, “Joint deep modeling of users and items using reviews for recommendation,” in *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining (WSDM ’17)*, 2017, pp. 425–434.
- [8] P. Massa and P. Avesani, “Trust-aware recommender systems,” in *Proceedings of the 2007 ACM Conference on Recommender Systems (RecSys ’07)*, 2007, pp. 17–24.
- [9] Document GraphRAG, “Graph-based retrieval-augmented generation for industrial document question answering,” 2024.

- [10] Y. Yao, D. Ye, P. Li, X. Han, Y. Lin, Z. Liu, Z. Liu, L. Huang, J. Zhou, and M. Sun, “DocRED: A large-scale document-level relation extraction dataset,” in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 764–777. [Online]. Available: <https://aclanthology.org/P19-1074>
- [11] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, “SQuAD: 100,000+ questions for machine comprehension of text,” in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016, pp. 2383–2392. [Online]. Available: <https://arxiv.org/abs/1606.05250>
- [12] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning, “Hotpotqa: A dataset for diverse, explainable multi-hop question answering,” in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018, pp. 2369–2380. [Online]. Available: <https://arxiv.org/abs/1809.09600>
- [13] J. Müller, L. Schmitt, and P. Hofmann, “A design science based hybrid rag and knowledge graph integration for industrial applications,” *Electronics*, vol. 14, no. 4, pp. 567–582, 2025.
- [14] NVIDIA Research, “Facts: A framework for enterprise rag chatbots,” *arXiv preprint arXiv:2407.07858*, 2024.
- [15] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, “Dense passage retrieval for open-domain question answering,” in *EMNLP*, 2020, pp. 6769–6781.
- [16] P. Covington, J. Adams, and E. Sargin, “Deep neural networks for youtube recommendations,” in *RecSys*, 2016, pp. 191–198.
- [17] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, “A comprehensive survey on graph neural networks,” *IEEE Transactions on Neural Networks and Learning Systems*, 2020.

Appendix A: Presentation

E-COMMERCE CHATBOT WITH RAG-POWERED PRODUCT RECOMMENDATION

Final Presentation
6th March 2026

PRESENTED BY:
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2

Problem Definition

Existing e-commerce chatbots provide generic responses with limited explainability. Customers often struggle to find the right products efficiently, while static pre-trained models produce outdated or hallucinated recommendations.

3

Project Objective

- To develop a semantic retrieval mechanism using dense vector embeddings for improved product search relevance.
- To implement a dual-stage retrieval architecture consisting of:
 - Top-K similarity-based candidate generation
 - Large Language Model (LLM)-based pairwise reranking
- To generate graph-grounded explanations that justify recommendations based on matched specifications.
- To design an interactive user interface that supports natural language queries and personalized recommendation workflows.

4

Purpose and Need

To develop a semantic product retrieval system that captures user intent using embeddings and product relationship knowledge, enabling diverse, contextually relevant, and non-redundant recommendations.

Need:

- Keyword-based or similarity-only product search does not fully capture user intent.
- Existing retrieval methods lack structured knowledge of product relationships.
- User intent is often not understood, leading to irrelevant or missed product recommendations.
- Users require more accurate and diverse recommendations that reflect the true meaning of their queries.

5

Paper 1: Leveraging RAG With Transformer for Context-Based Personalized Recommendations[1]

Methodology:

- An instruction-tuned LLM retrieves and re-ranks preference contexts relevant to the user query.
- The Top-K preferences are converted into normalized embeddings.
- A transformer model processes these embeddings using multi-head attention.
- A linear layer with softmax predicts the Top-N recommended items.

Strengths:

- Effectively mitigates cold-start problem.
- Combines semantic retrieval with deep contextual modeling.
- Demonstrates improved recall and ranking accuracy.

6

PAPER 2: Document GraphRAG [2]

Methodology:

- Preprocess documents and convert them into text chunks with embeddings.
- Construct a Document Knowledge Graph (DKG) capturing relationships between document entities and sections.
- Use the graph to link related document fragments and improve retrieval relevance.
- Retrieve context using graph-guided retrieval + vector similarity search.
- Provide structured context to an LLM to generate final answers.

Strengths:

- Multi-hop contextual reasoning
- Structured knowledge grounding
- Reduced hallucination

7

PAPER 3: Retrieval augmented generation for 10 large language models and its generalizability in assessing medical fitness [3]

Methodology :

- Collect medical guideline datasets (local and international clinical guidelines).
- Convert guidelines into retrievable knowledge sources for the RAG pipeline.
- Process clinical queries and retrieve relevant guideline sections.
- Provide retrieved context to multiple LLMs (GPT-4, Gemini, LLaMA) for response generation.
- Evaluate performance using accuracy, safety, hallucination rate, and consistency metrics.

Strengths:

- Handles sparse and cold-start scenarios
- Improves accuracy over GNN-only baselines
- Demonstrates strong generalizability

8

PAPER 4: FACTS About Building Retrieval Augmented Generation-based Chatbots [4]

Methodology :

- Collect enterprise data from multiple sources (SharePoint, Slack, documents, PDFs).
- Perform document preprocessing, chunking, and embedding generation.
- Store embeddings in a vector database for semantic retrieval.
- Process user queries using query reformulation, metadata enrichment, and reranking.
- Generate answers with LLM using retrieved enterprise documents as context.

Strengths:

- Practical deployment insights
- Emphasis on scalability and monitoring
- Cost-performance tradeoff analysis

9

Research Gap Identification

- Existing approaches lack multi-stage retrieval pipelines (retrieval → filtering → ranking → reranking).
- Most RAG systems focus on document QA, not product recommendation.
- Limited support for user constraints and product attributes (price, specifications, preferences).
- Many recommendation frameworks require heavy model training, reducing deployability.
- Current RAG pipelines often suffer from retrieval noise and context fragmentation.
- Most systems do not provide explainable recommendations.

10

How Our System Addresses These Gaps

- Uses RAG for semantic product retrieval.
- Implements multi-stage ranking(Vector Search → Filtering → Ranking → LLM Reranking).
- Supports constraint-aware recommendations (budget, features, intent).
- Integrates knowledge graph reasoning for explainability.
- Generates interpretable recommendations with explanations.

11

Proposed Methodology

Query Understanding and Category Detection

- When a user submits a natural language query, it is analyzed to detect the product category.
- Category detection ensures that retrieval is restricted to relevant product datasets, this helps in improving search efficiency.

Semantic Retrieval using Embeddings

- The user query is converted into a vector embedding using a sentence-embedding model.
- Product descriptions stored in the database are also pre-encoded as embeddings.
- Cosine similarity is computed between the query embedding and product embeddings and the system retrieves the Top-K most semantically similar products.

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Deterministic Constraint Filtering

- Product metadata such as price, brand, storage, RAM, power rating, etc. is checked against the constraints inferred from the query.
- Products that do not satisfy required constraints are removed.

Hybrid Score Fusion

- The remaining candidates, a hybrid score is computed by combining Semantic similarity score and Specification matching score.
- Score normalization is applied so that both scores contribute fairly to the ranking. This produces a balanced relevance score for each product.

LLM-Based Reranking

- The filtered candidate products are sent to Phi-3 (via Ollama) in a single listwise prompt, where the model evaluates all candidates together relative to the user query.

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Knowledge Graph Integration for Explainability

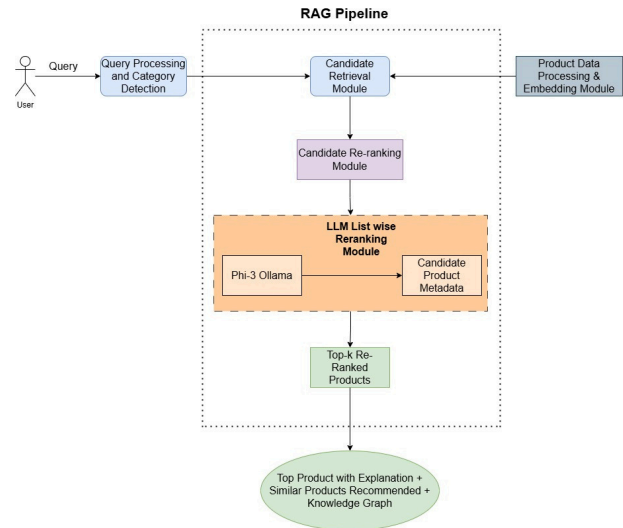
- The system uses a Neo4j knowledge graph that stores relationships such as: product-> category, product->brand, etc.
- These relationships are used to generate explanations for recommendations.

Final Recommendation Generation

- The system outputs the Top-N ranked products, along with : Product details, Relevance ranking, Knowledge-graph-based explanations.
- This makes the final recommendations accurate, constraint-validated, and explainable.

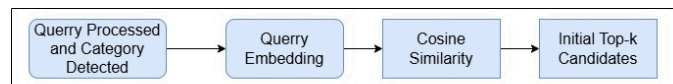
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Architecture Diagram

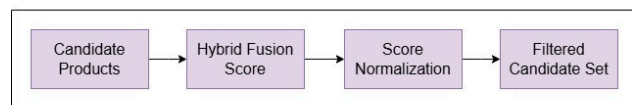


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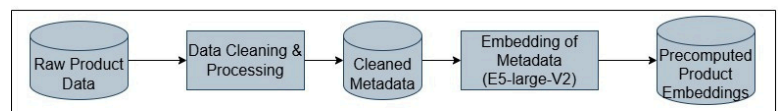
Architecture Diagram



Candidate Retrieval Module



Candidate Re-ranking Module

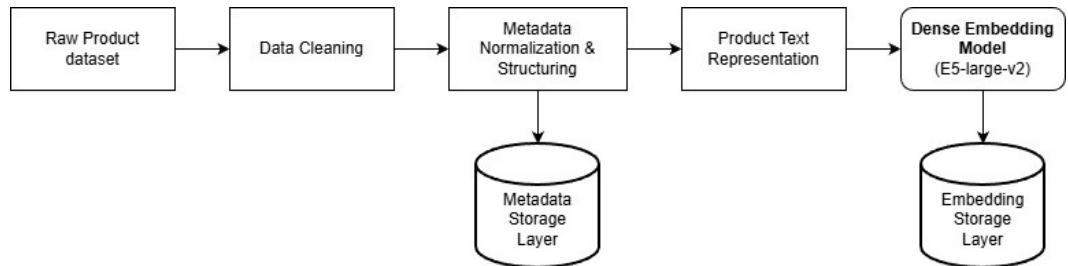


Product Data Processing & Embedding Module

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MODULE WISE DIAGRAMS

Preprocessing module

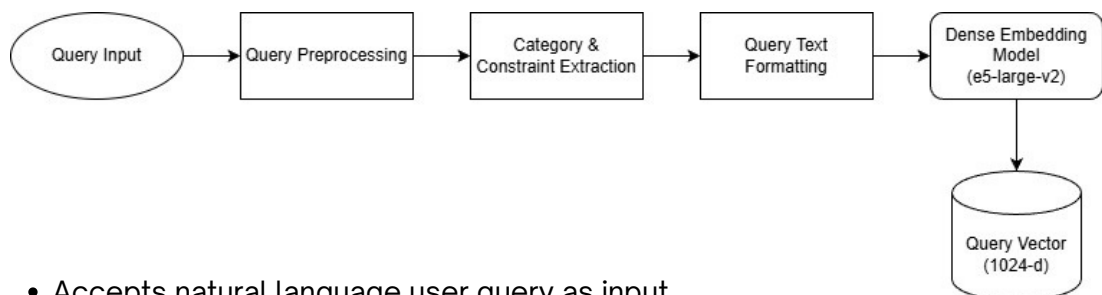


- Performs offline preprocessing of raw product dataset
- Cleans and normalizes metadata (price, brand, specs, text fields)
- Generates unified product text representation for embedding
- Uses E5-large-v2 to convert product text into dense vectors
- Stores structured metadata in Metadata Storage Layer
- Stores product embeddings in Embedding Storage Layer for fast retrieval

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MODULE WISE DIAGRAMS

Query Embedding module

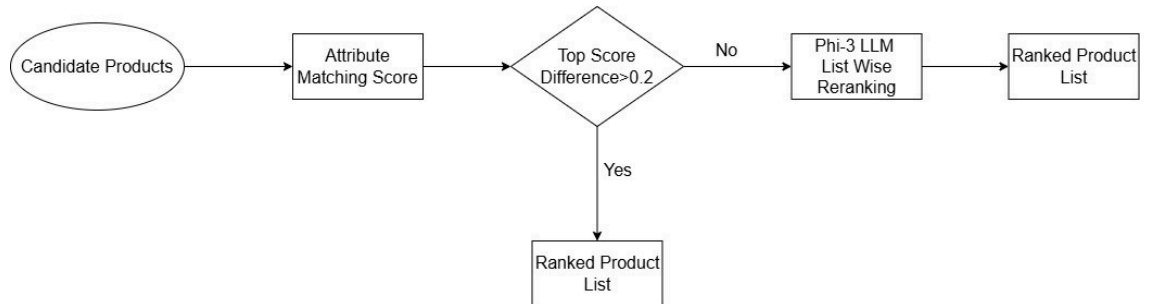


- Accepts natural language user query as input
- Performs preprocessing of the query
- Extracts product category and user constraints (price, brand, specs)
- Formats query using E5 instruction template (“query: ...”)
- Encodes query into a 1024-dimensional dense vector using E5-large-v2
- Outputs query vector for similarity-based retrieval

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MODULE WISE DIAGRAMS

Reranking module



- Receives filtered candidate products.
- Extracts relevant product metadata such as descriptions and attributes.
- Evaluates ranking confidence using the score difference between top candidates, if low undergoes reranking
- Outputs top-N reranked products for explanation and recommendation display.

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RESULTS AND OUTPUTS

```
=====
Query: samsung tv under 50000
GT size: 26
Top 10 results: [40, 41, 304, 286, 39, 35, 305, 374, 245, 37]

@10
Recall: 0.3846
NDCG : 1.0000

@20
Recall: 0.7308
NDCG : 0.9652

@40
Recall: 0.9615
NDCG : 0.9707
```

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RESULTS AND OUTPUTS

```
=====
Query: mixer-grinder under 1000
GT size: 89
Top 10 results: [56, 112, 9, 133, 25, 153, 38, 204, 135, 83]

@10
Recall: 0.1124
NDCG : 1.0000

@20
Recall: 0.2135
NDCG : 0.9604

@40
Recall: 0.4045
NDCG : 0.9198
```

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RESULTS AND OUTPUTS

```
=====
Query: iphone
Top 10 results: [112, 175, 308, 34, 16, 347, 137, 7, 150, 440]

@10
Recall: 0.3333
NDCG : 0.9266

@20
Recall: 0.6667
NDCG : 0.9179

@40
Recall: 0.9259
NDCG : 0.9324
```

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RESULTS AND OUTPUTS

Query: Recommend me an android smartphone

Detected Category: SMARTPHONES

★ Best Recommendation

Redmi Note 9 Pro (Interstellar Black, 4GB RAM, 64GB Storage)- Latest 8nm Snapdragon 720G & Alexa Hands-Free Capable

Price unavailable

4GB RAM

💡 Other Recommended Products

OPPO A31 (Mystery Black, 6GB RAM, 128GB Storage) with No Cost EMI/Additional Exchange Offers

₹12,490

6GB RAM

OPPO A74 5G (Fluid Black, 6GB RAM, 128GB Storage) with No Cost EMI/Additional Exchange Offers

₹15,490

6GB RAM

5G connectivity

OPPO A74 5G (Fantastic Purple, 6GB RAM, 128GB Storage) with No Cost EMI/Additional Exchange Offers

₹15,490

6GB RAM

5G connectivity

IQOO Z6 Pro 5G by vivo (Phantom Dusk, 6GB RAM, 128GB Storage) | Snapdragon 778G 5G | 66W FlashCharge | 1300 nits Peak Brig...

₹22,999

6GB RAM

5G connectivity

66W power

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RESULTS AND OUTPUTS

💡 Why this recommendation?

Matches your search for 'Recommend me a android smartphone'

This product was recommended because it closely aligns with the requirements expressed in your search for 'Recommend me a android smartphone'. Within the smartphones category, users typically look for products that provide a balanced combination of performance, reliability, and useful specifications. Based on the available product data, this item includes features such as relevant product specifications. These specifications play an important role in determining how well a device performs in everyday use. For example, storage capacity and hardware configuration influence how quickly applications load, how smoothly tasks run, and how effectively the system handles multitasking or demanding workloads. In addition, the product description and metadata show strong alignment with the intent expressed in your search query. Products that contain relevant specifications and closely match the requested features are ranked higher within the recommendation results. Because this item demonstrates strong relevance and contains attributes commonly associated with high-quality smartphones products, it was identified as a suitable and reliable option for your needs.

📊 Knowledge Graph

The Knowledge Graph visualization shows a central node labeled 'User Query' connected to several product nodes. The products listed are: #1Redmi Note 9 Pro (Interstellar Black, 4G..., #2realme jaro 5G (Spec) Black, 4GB RAM-64..., #3realme narzo 5G (Spec) Black, 6GB RAM-12..., #3OPPO A74 5G (Fluid Black, 6GB RAM, 128GB..., #4OPPO A73s (Razzy White, 4GB, 128GB Stora..., #2OPPO A74 5G (Fantastic Purple, 6GB RAM, 12..., #5OPPO A31 (Mystery Black, 6GB RAM, 128GB..., and #9IQOO Z6 Pro 5G by vivo (Phantom Dusk, 6G... The graph is plotted on a 2D coordinate system with axes labeled x and y.

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RESULTS AND OUTPUTS

Query: Recommend me a gaming laptop

Detected Category: LAPTOPS

★ Best Recommendation

HP Omen 15-AX205na Gaming 15.6"

₹98,910

8GB RAM

NVIDIA graphics

128GB SSD

💡 Other Recommended Products

HP Omen 17-W006na Gaming 17.3"

₹125,910

8GB RAM

NVIDIA graphics

128GB SSD

Dell Alienware 15 Gaming 15.6"

₹119,610

16GB RAM

NVIDIA graphics

256GB SSD

MSI GT72S Dominator Gaming 17.3"

₹139,107

16GB RAM

NVIDIA graphics

256GB SSD

MSI GT72VR Dominator Gaming 17.3"

₹224,910

16GB RAM

NVIDIA graphics

256GB SSD

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RESULTS AND OUTPUTS

💡 Why this recommendation?

Matches your search for 'Recommend me a gaming laptop'

Also offers 128GB SSD storage

Also offers 15.6" display

This product was recommended because it closely aligns with the requirements expressed in your search for 'Recommend me a gaming laptop'. Within the laptops category, users typically look for products that provide a balanced combination of performance, reliability, and useful specifications. Based on the available product data, this item includes features such as 128GB SSD storage, 15.6" display. These specifications play an important role in determining how well a device performs in everyday use. For example, storage capacity and hardware configuration influence how quickly applications load, how smoothly tasks run, and how effectively the system handles multitasking or demanding workloads. In addition, the product description and metadata show strong alignment with the intent expressed in your search query. Products that contain relevant specifications and closely match the requested features are ranked higher within the recommendation results. Because this item demonstrates strong relevance and contains attributes commonly associated with high-quality laptops products, it was identified as a suitable and reliable option for your needs.

📊 Knowledge Graph

The Knowledge Graph visualization shows a central node labeled 'User Query' connected to several product nodes. The products are: HP Omen 15-AX205na Gaming 15.6", HP Omen 17-W006na Gaming 17.3", Dell Alienware 15 Gaming 15.6", MSI GT72S Dominator Gaming 17.3", and MSI GT72VR Dominator Gaming 17.3". The graph also includes feature nodes such as '8GB RAM', '16GB RAM', '128GB SSD', '256GB SSD', 'NVIDIA graphics', and '15.6" display'. The connections indicate that the recommended product (HP Omen 15-AX205na) is highly relevant to the user query and matches the specified features.

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ASSUMPTIONS

- The Amazon product dataset used for training and retrieval is accurate, complete, and periodically updated.
- Product metadata (price, specifications, brand, category) is correctly structured and consistent.
- Users provide meaningful natural language queries containing sufficient context or constraints.
- Constraint information such as RAM, budget, or brand can be reliably extracted from user queries.
- Neo4j graph correctly retrieves structured product–attribute relationships.
- Internet connectivity is stable for model loading (Ollama), operations.
- User preference data stored in Supabase is secure and accessible without data corruption.

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Work Breakdown and Responsibilities

- Cleaning and extraction of dataset - Visal and Rohit
- Embedding and Indexing of the products from datasets - Sneha and Nikitha
- Implementation of Reranking - Visal and Rohit
- Explainability - Visal and Rohit
- Product Knowledge Graph - Visal and Rohit
- User Profile and Interface - Nikitha and Sneha
- Documentation - Sneha and Nikitha

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Hardware Requirements

- Development Machine:
 - Minimum 8 GB RAM
 - Multi-core CPU (Intel i5 / Ryzen 5 or higher)
- GPU (Recommended for Embedding & LLM inference):
 - NVIDIA GPU (CUDA supported)
 - Or Cloud GPU (Google Colab / Kaggle / Cloud VM)
- Storage:
 - Minimum 20 GB free space for dataset, embeddings, and logs
- Stable Internet Connection:
 - Required for model downloads, database access, and API connectivity

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Software Requirements

- Programming Language
 - Python 3.9
- Frameworks & Libraries
 - Sentence-Transformers (E5-large-v2 embeddings)
 - NumPy, Pandas
 - Streamlit (Chatbot UI)
 - Ollama (Phi-3 LLM integration)
 - Neo4j Aura(Product Knowledge Graph)
 - Plotly & NetworkX (Graph visualization)
- Databases
 - Neo4j Graph Database

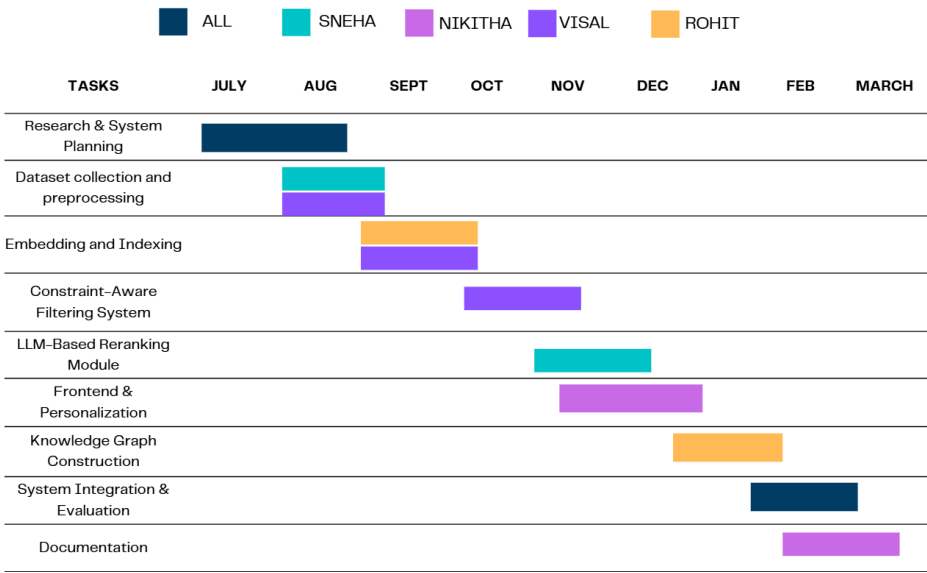
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Dataset Used

Amazon Product Dataset

- Product titles
- Descriptions
- Specifications
- Brand & Category information
- Price metadata

Gantt Chart



RISKS AND CHALLENGES

- Reasoning Hallucination Risk
 - Even with retrieval grounding, the LLM-based reranking module may generate partially inaccurate contextual reasoning if prompts are not well-structured.
- Retrieval Latency
 - Large embedding stores and cosine similarity computations may increase response time, affecting real-time conversational performance.
- Item Cold Start Scenarios
 - New products with limited metadata may not be optimally ranked due to sparse descriptive information.

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RISKS AND CHALLENGES

- Graph Consistency Issues
 - Incomplete or inconsistent product–attribute relationships may impact explanation accuracy.
- Constraint Extraction Errors
 - Incorrect parsing of user constraints (e.g., RAM, budget) may lead to filtering inaccuracies.
- Dependency Risks
 - Changes in external libraries (Ollama, sentence-transformers, Neo4j drivers) may affect compatibility.

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Conclusion

- A semantic product retrieval system was developed to understand user's intent.
- The system converts user queries and products into vector embeddings, enabling retrieval based on semantic similarity.
- By combining embedding-based retrieval with product-relationship knowledge, the system provides more relevant and diverse recommendations.
- The approach reduces redundant results and improves the overall quality of retrieved products.
- The proposed framework demonstrates how semantic search and structured product knowledge can

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References

- [1] Y. Zhang, X. Wang, and H. Li, "Leveraging Retrieval-Augmented Generation with Transformer for Context-Based Personalized Recommendations," IEEE Access, vol. 13, pp. 12674–12685, 2025.
- [2] Document GraphRAG, "Graph-Based Retrieval-Augmented Generation for Industrial Document Question Answering," 2024.
- [3] M. I. Razzak et al., "Retrieval-augmented generation for 10 large language models and its generalizability in assessing medical fitness," npj Digital Medicine, vol. 8, no. 1, pp. 45–60, 2025.
- [4] NVIDIA, "FACTS About Building Retrieval Augmented Generation-Based Chatbots," 2024.
- [5] P. Massa and P. Avesani, "Trust-aware recommender systems," in Proc. ACM Conf. Recommender Systems (RecSys '07), 2007, pp. 17–24.
- [6] G. Zheng, V. Noroozi, and P. S. Yu, "Joint deep modeling of users and items using reviews for recommendation," in Proc. ACM Int. Conf. Web Search and Data Mining (WSDM '17), 2017, pp. 425–434.
- [7] W. X. Zhao et al., "A survey of large language models," arXiv preprint arXiv:2303.18223, 2023.
- [8] J. Johnson, M. Douze, and H. Jégou, "Faiss: A library for efficient similarity search and clustering of dense vectors," in Proc. IEEE ICASSP, 2017, pp. 2617–2621.
- [9] Z. Xiao et al., "BGE embeddings: Efficient and effective text embedding models," 2023. [Online]. Available: <https://huggingface.co/BAAI/bge-large-en-v1.5>
- [10] Y. Yao et al., "DocRED: A large-scale document-level relation extraction dataset," in Proc. ACL, 2019, pp. 764–777.

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Thank You

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skills, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions for complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design or develop systems, components, or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and the World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

PO8: Individual and Collaborative Teamwork: Function effectively as an individual and as a member or leader in diverse or multidisciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, and make effective presentations considering cultural,

language and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for and have the preparation and ability for (i) independent and life-long learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real-world problems by applying knowledge across

domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative or creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C:

CO and Justifications

COURSE OUTCOMES:

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

**MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND
COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)**

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

MAPPING	LOW/ MEDIUM / HIGH	JUSTIFICATION
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.

101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.

101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.

101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.

101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.

101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.